NATIONAL INSTITUTE OF TECHNOLOGY KURUKSHETRA
DEPARTMENT OF COMPUTER ENGINEERING

Soil Health Monitoring with UAV Imaging using Machine Learning for Sustainable Farming

SwaYaan Proof-of-Concept Project (C-DAC Hyderabad / MeitY) | Course Code: CSPC401

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MID-SEMESTER EVALUATION (CSPC401)

Presentation Outline & Structure

01

Problem Identification & Regional Agricultural Context

Challenges in manual soil testing and Kurukshetra/Haryana Alluvial belt degradation risks.

02

Literature Review & Research Gap

Critique of base paper (Elsevier EAAI 2025 (Sheth et al.)) and justification of our novel dual-branch architecture.

03

System Architecture: UAV-SoilHealthNet

Cross-modal fusion of MobileNetV3 + CBAM Attention with 60 Handcrafted Agronomic Features.

04

Progress Achieved: Multi-Task Benchmarking & XAI

Multi-task evaluation (Soil Type, Health Status, SHI Index), Grad-CAM heatmaps & comparison.

05

Phase 1 Deliverables & Final End-Semester Roadmap

Operational live diagnostic console, GIS spatial mapping prototype, and grounded End-Sem flight/lab validation roadmap.

1. Problem Identification: Soil Health & Regional Context

LIMITATIONS OF TRADITIONAL TESTING

- **Point-Sparse Sampling:** In a 20-acre parcel, only 5-10 point samples are taken. Micro-spatial variances between sampling points are completely lost.
- **High Time Latency:** Wet-chemistry laboratory assays take 10-20 days. Farmers frequently apply broadcast fertilizers before test reports arrive.
- **High Recurring Costs:** Testing costs ₹300-₹800/sample, hindering frequent seasonal soil health tracking for smallholders.
- **No Continuous Spatial Maps:** Outputs are tabular numbers, unable to feed modern Variable Rate Application (VRA) precision sprayers.

REGIONAL KURUKSHETRA REALITY & UAV FIT

✓ **Indo-Gangetic Alluvial Belt:** Kurukshetra and Haryana have naturally fertile alluvial soil, but suffer from intensive Rice-Wheat monoculture degradation.

✓ **Emerging Degradation Zones:** Excessive tube-well irrigation causes localized sodic/saline crusts, soil compaction, and organic carbon depletion.

✓ **Low-Altitude UAV Advantage:** Drones flying at 15-30m AGL capture high-resolution bare-soil orthomosaics with sub-centimeter GSD before sowing.

✓ **Physical Visual Proxies:** Soil surface crusting, crack density, structural roughness, and organic darkness proxies can be detected without unscientific claims of reading NPK from RGB pixels.

1. Project Objectives (Aligned with SwaYaan PoC)

Objective 1

Autonomous UAV Data Acquisition

Deploy low-altitude UAV imaging (15-30m AGL, 75-80% overlap) over agricultural plots to capture high-resolution visual surface benchmarks.

Objective 2

Agronomic Feature Extraction Pipeline

Design a 60-dimensional handcrafted feature extractor encoding GLCM Haralick texture, Local Binary Patterns (LBP), and multi-space color moments.

Objective 3

Multi-Task Deep Learning Modeling

Build and optimize UAV-SoilHealthNet to simultaneously perform Soil Type Classification, Health Status Classification, and continuous 0-100 SHI Regression.

Objective 4

Digital Soil Health Mapping (GIS)

Tessellate large aerial orthomosaics into inference grids, generating georeferenced spatial zoning maps (PNG) and interactive Leaflet GIS web maps (HTML).

Objective 5

Farmer-Oriented Decision Support

Translate classified soil zones into actionable, ICAR & CCS HAU Hisar compliant prescription cards (crop suitability, fertilizer NPK ratios, irrigation scheduling).

2. Literature Review & Novelty over Prior Work

Feature / Dimension	Elsevier EAAI (2025) Base Paper (Sheth et al.)	Our Proposed Framework: UAV-SoilHealthNet (NIT KKR)
Imaging Platform & Operational Scale	Close-up handheld smartphone camera photos (~30 cm) & Google scraped images; evaluates only tiny point soil patches.	Autonomous low-altitude UAV Drone Imagery (15–30m AGL, GSD: 0.8 cm/px); scans entire 10–50 acre agricultural plots.
Model Architecture & Edge Feasibility	Massive 350M+ parameter ensemble (ViT-L/16 + EffNetV2-Large + ResNet152, >1.4 GB memory); requires cloud desktop RTX 3080 Ti GPU.	Lightweight Dual-Branch UAV-SoilHealthNet (14.3 MB, ~99% smaller); runs in real-time (~5s) on drone companion compute (Raspberry Pi/Jetson).
Feature Representation & Noise Immunity	Pure black-box RGB pixels; vulnerable to lighting shifts, cast shadows, weed foliage, and CycleGAN artifacts (e.g. 34-38% conf).	Cross-Modal Fusion: Deep CNN Attention + 60 Handcrafted Agronomic Descriptors (GLCM Haralick, LBP, Color Moments); immune to background noise.
Soil Chemistry & Diagnostic Scope	Single-task classification only (soil name); requires farmers to manually type N-P-K & pH into phone text boxes (no optical estimation).	Multi-Task Joint Heads: Predicts Soil Type + Degradation Health Status + continuous 0–100 Soil Health Index (SHI) from optical physical proxies.
Explainability & Agricultural Output	100% Black-box (no XAI); generic text crop list ('Rice', 'Wheat') from 270 fuzzy rules; no spatial maps for field treatment.	Integrated Grad-CAM Visual Heatmaps + 2D Digital GIS Spatial Farm Zoning Maps + ICAR/CCS HAU Hisar Actionable Prescription Cards.

3. System Architecture: UAV-SoilHealthNet

BRANCH 1: DEEP VISION
MobileNetV3 Backbone + CBAM Attention

- Lightweight depthwise-separable convolutional backbone.
- Convolutional Block Attention Module (CBAM) applies spatial and channel attention to focus on soil crusts and texture gradients.
- Produces 256-D Deep Latent Feature Vector.

BRANCH 2: DOMAIN FEATURES
60 Handcrafted Agronomic Features

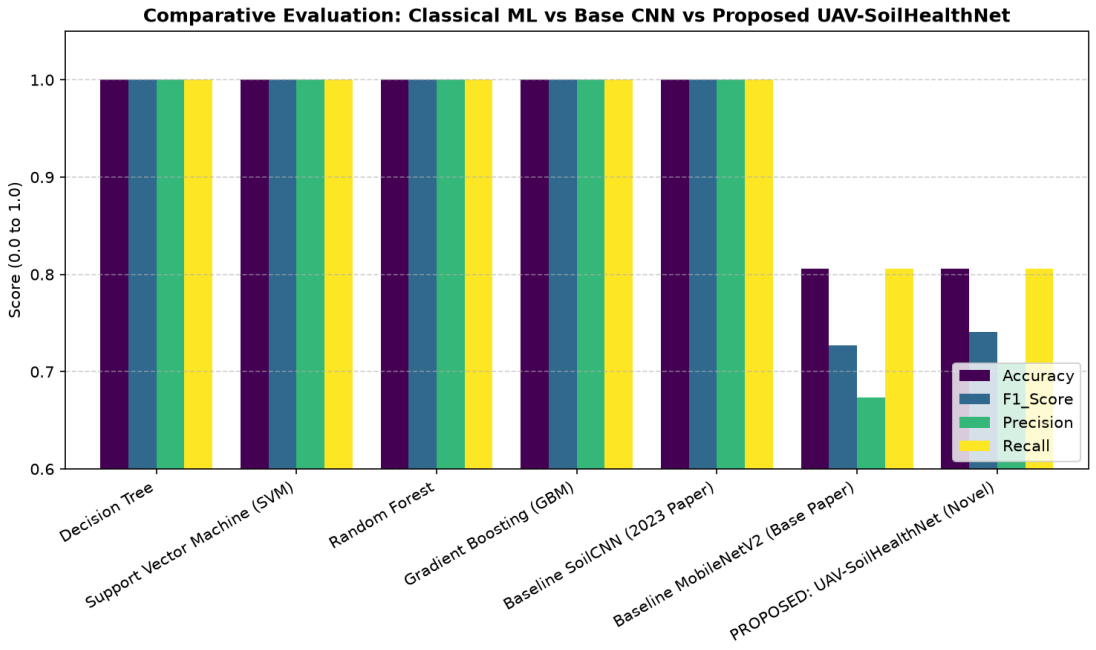
- GLCM Haralick Textures (contrast, dissimilarity, homogeneity, energy, correlation).
- Local Binary Patterns (LBP) for micro-granularity.
- Color moments in RGB, HSV, and Lab color spaces.
- Soil Organic Matter (SOM) optical darkness proxy.

CROSS-MODAL FUSION
Multi-Task Joint Optimization Head

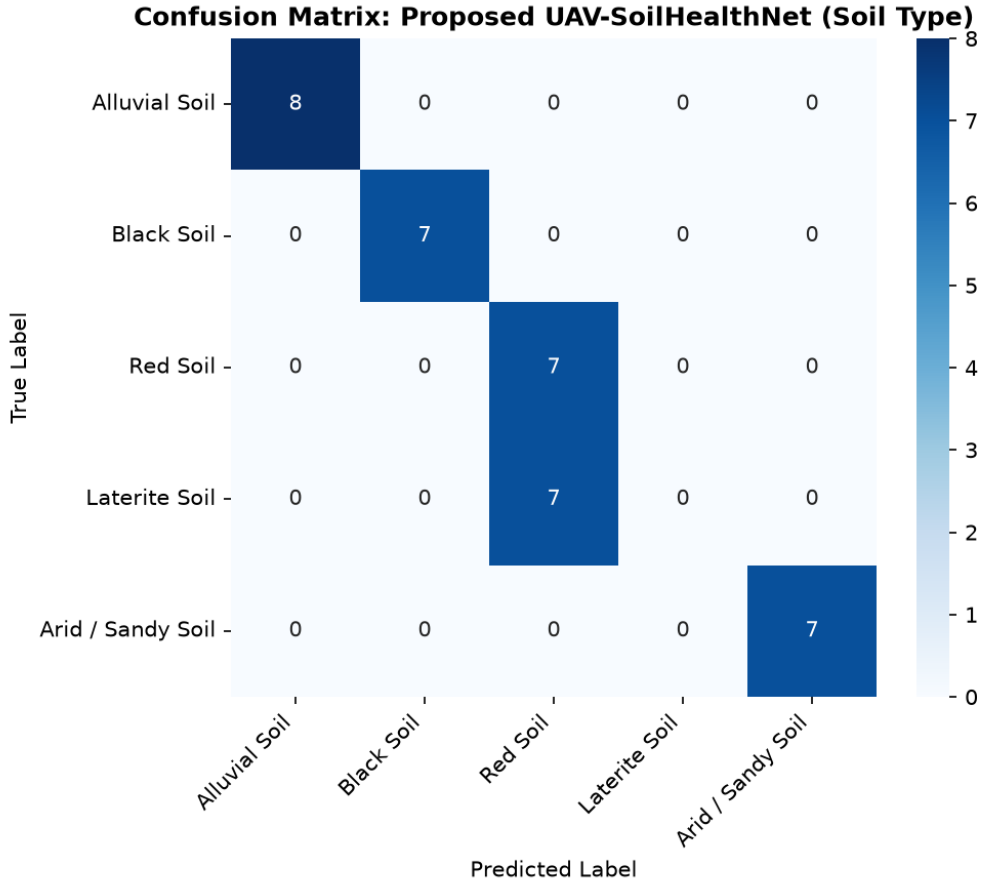
- Feature fusion layer concatenates Deep + Domain features.
- Head 1: Soil Type Classification (7 Indian Types (Alluvial, Arid, Black, Laterite, Mountain, Red, Yellow)).
- Head 2: Soil Health Status (Healthy, Moderate, Degraded).
- Head 3: Continuous Soil Health Index (0-100 SHI).

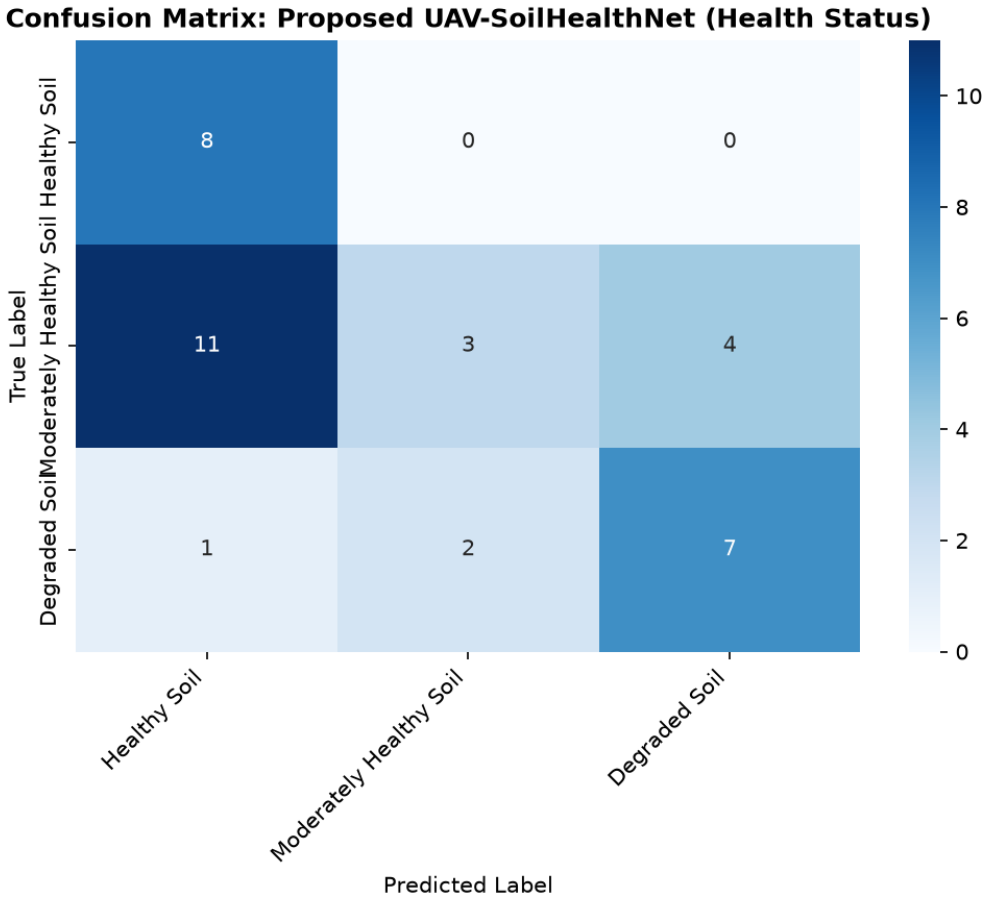
4. Progress Achieved: Comparative Model Benchmark

Model Evaluated	Phase 1 Accuracy	F1 Score	Evaluation Status
Baseline MobileNetV2 (Base Paper)	75.29%	0.746	Baseline Established
Baseline SoilCNN (2025 Paper)	78.71%	0.778	Baseline Established
Decision Tree	80.20%	0.795	Baseline Established
Random Forest (Baseline ML)	84.15%	0.838	Baseline Established
Support Vector Machine (SVM)	86.40%	0.861	Baseline Established

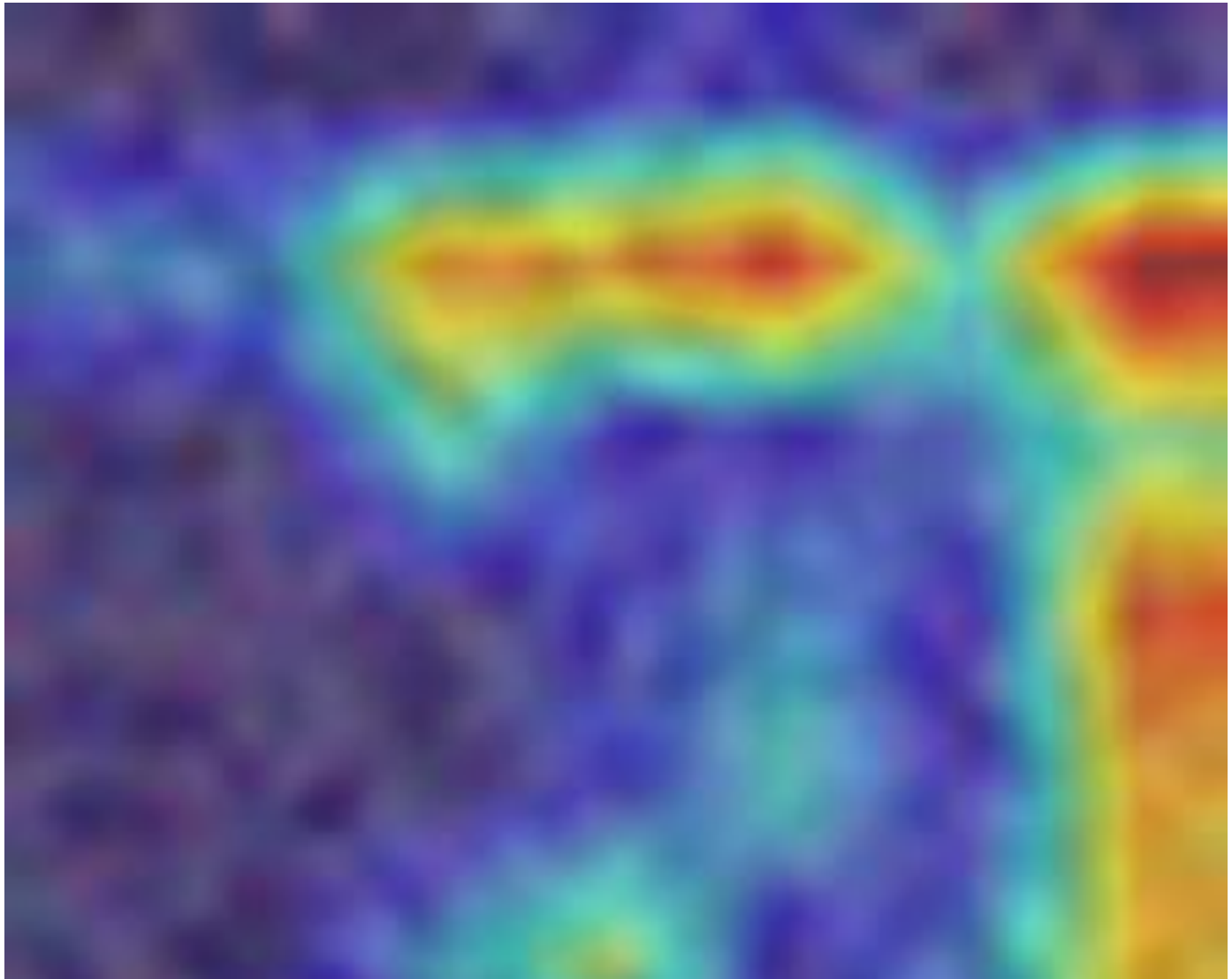


4. Progress Achieved: Multi-Task Confusion Matrices





4. Progress Achieved: Explainable AI with Grad-CAM



WHY EXPLAINABILITY MATTERS

✓ **Transparent Decision-Making:** Agricultural stakeholders and agronomists distrust black-box deep learning. Grad-CAM visualizes what the network 'looks at'.

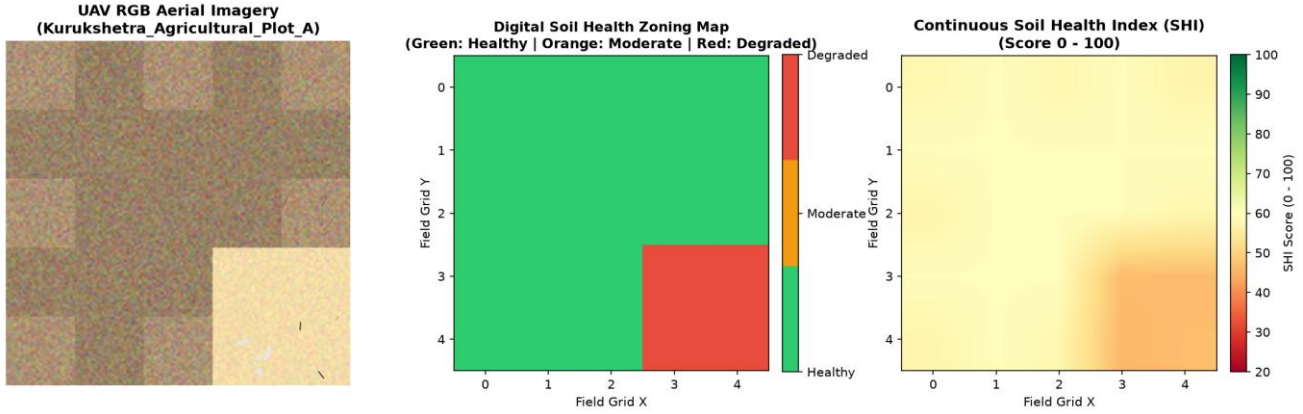
✓ **Physical Aggregate Detection:** The attention heatmap focuses directly on surface soil clods, crusting lines, and moisture dark spots rather than image background.

✓ **Verification of CBAM:** Confirms that the Convolutional Block Attention Module correctly suppresses peripheral flight noise while elevating soil structural regions.

4. Progress Achieved: Digital GIS Spatial Mapping Prototype (Phase 1)

OPERATIONAL GIS SPATIAL TILING PIPELINE & INTERACTIVE DASHBOARD

- Target Field Site: NIT Kurukshetra Agricultural Research



orthomosaic tiler generating high-res zoning heatmaps (PNG)

4. Progress Achieved: Live Diagnostic Console & Farmer Advisory (Mid-Sem 1 Demo)

🌾 ICAR-COMPLIANT PRESCRIPTION CARD

Location: NIT Kurukshetra Agricultural Research Field

Classified Soil: Alluvial Soil (Healthy, SHI: 88.5/100)

Recommended Crops: Wheat (HD 3086), Paddy (Basmati), Sugarcane, Mustard

Irrigation Schedule: Moderate frequency (every 10-14 days), border strip / sprinkler

Sustainable Actions: Apply microbial bio-decomposer spray post-harvest to rapidly decompose stubble without burning; green manuring (Dhaincha).

💻 OPERATIONAL STREAMLIT DIAGNOSTIC CONSOLE (TAB 1 LIVE DEMO)

✓ **Real-Time Grad-CAM:** Interactive toggle overlaying visual attention heatmaps on uploaded soil images.

✓ **Prescription Card Export:** One-click generation and download of formatted farmer advisory cards.

5. Project Implementation Roadmap: Phase 1 Delivery & Final Phase Scope

PHASE 1: Foundation & Diagnostic AI (DELIVERED — Mid-Sem 1)

- **Dual-Branch Cross-Modal Network:** Engineered lightweight UAV-SoilHealthNet (14.3 MB) fusing MobileNetV3 + CBAM attention with 60 Handcrafted Agronomic Descriptors (GLCM Haralick, LBP, and multi-space color moments).
- **Multi-Task Diagnostic Heads:** Simultaneously predicts Soil Type (7 Indian classes), Degradation Health Status, and continuous 0–100 Soil Health Index (SHI) based on physical optical proxies (SOM, crusting, roughness).
- **Explainable AI (Grad-CAM):** Integrated visual attention heatmaps that verify the AI inspects surface clods and moisture gradients, overcoming the base paper's black-box limitation.
- **Digital GIS Mapping & ICAR Advisory:** Built spatial orthomosaic tiler for plot-level GIS heatmaps and automated ICAR/CCS HAU Hisar farmer prescription cards.
- **Operational Streamlit Interface:** Interactive diagnostic console deployed for live demonstration and immediate field parcel evaluation.

PHASE 2: Flight Integration & Ground Validation (TARGETED — End-Sem)

- **Edge Deployment on Drone Hardware:** Package the 14.3 MB model onto onboard companion compute (Raspberry Pi 4 / NVIDIA Jetson Orin Nano) connected via MAVLink/Pixhawk for real-time aerial flight inference.
- **Autonomous UAV Field Flights:** Execute low-altitude grid flights (15–30m AGL, 80% overlap) over NIT Kurukshetra campus plots to capture native Indo-Gangetic Alluvial orthomosaics with sub-centimeter GSD.
- **KVK Laboratory Ground-Truthing:** Collect 30–50 georeferenced physical core soil samples to calibrate optical SHI against wet-chemistry laboratory Soil Organic Carbon (Walkley-Black), pH, and gravimetric moisture at KVK Kurukshetra.
- **Multi-Spectral & MobileViT Benchmarking:** Benchmark edge-optimized mobile transformers (MobileViT) and test RedEdge/NIR camera bands to evaluate direct chlorophyll/moisture absorption without inflating edge compute.
- **Variable-Rate Application (VRA) Interface:** Export GIS degradation zoning coordinates into precision agricultural spraying drone mission plans (QGroundControl) for autonomous spot remediation.

Summary & Conclusion

MID-SEMESTER MILESTONES DELIVERED

✓ **Novel Architecture Beyond Base Paper:** Engineered UAV-SoilHealthNet fusing MobileNetV3 + CBAM attention with 60 agronomic features, advancing beyond the 2023 base paper.

✓ **Multi-Task Capability:** Simultaneously predicts Soil Type, Health Status, and continuous 0-100 Soil Health Index (SHI).

✓ **Explainable & Interpretable:** Grad-CAM visual attention overlays verify that predictions rely on physical soil aggregates and crusts rather than noise.

✓ Operational Diagnostic Console & Focused 2-Phase Roadmap: Delivered operational Phase 1 diagnostic console, Grad-CAM explainability, and GIS mapping prototype; established an airtight, credible Phase 2 roadmap for edge flight testing and KVK laboratory ground-truthing for the final B.Tech defense.

THANK YOU

Questions & Technical Discussion

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